



WishTree

DEVELOPMENT BOARD

Quick IoT prototyping on
WiFi, Bluetooth, BLE, GSM, LoRa
on a single board.



More information and examples at :
www.wisethingz.com/wishtree



Runs
ESP8266
& ESP32

Both ESP8266 & ESP32
supplied- no soldering
required

Accommodates peripherals
directly on the board- no
breadboard needed

Programmable using
Arduino & other IDE's

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Available in
India | USA | Canada | UAE

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and distribution opportunities

- The receiver gets data from all nodes one by one and it updates readings of any node after set time period.

When RTC is turned on for the first time, its current date and time have to be set using `rtc.adjust ( )` function (given in the program with comment). Once date and time are set for RTC, we have to again comment out the `rtc.adjust ( )` function from program and upload the program again. So, the receiver program is to be uploaded two times in Arduino Nano board.

EFY Note

The source code
of this project
is available
for free download
at source.efymag.com

PARTS LIST

Semiconductors:

Board1, Board2 - Arduino Nano board
LED1, LED2 - 5mm LED

Resistors (all 1/4-watt, ±5% carbon):

R1, R4 - 470-ohm
R2, R3 - 10-kilo-ohm

Miscellaneous:

CON1 - 2-pin connector
CON2, CON3 - 4-pin connector
CON4 - 6-pin connector
SS1 - Soil sensor module
DHT11 - Temperature and humidity sensor

LDR1 - Light dependent resistor
RTC1 - DS1307/DS3231 RTC module

TX1 - 433MHz RF transmitter module

RX1 - 433MHz RF receiver module

ANT.1, ANT.2 - 1mm dia single-core, 35cm long copper wire
- Jumper wires
- 9V battery or regulated power supply

LabtestRx.ino code in Board2 is used to display the time, date, and status of nodes transmission on serial monitor. You need to include the related libraries *DHT.zip*, *RTCLib.zip*, and *VirtualWire.zip* before uploading the source codes.

Construction and testing

An actual-size PCB layout for the transmitter (sensor node) is shown in Fig. 4 and its components layout in Fig. 5. After assembling the circuit on PCB, connect 9V DC to CON1.

An actual-size PCB layout for the receiver is shown in Fig. 6 and its components layout in Fig. 7. After assembling the circuit on PCB, connect the board to a laptop or desktop computer to monitor the data sent from various nodes on serial monitor in Arduino IDE. **EFY**

There are two source codes: *Labtesttxnode.ino* code for sensor node and *LabtestRx.ino* code for receiver. The *Labtesttxnode.ino* code in Board1 is used to collect data from the sensors and process the data. The



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